

Impact of RBI Monetary Policy on Sectoral Stock Returns in India

The Project Report

submitted by

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under the supervision and guidance of

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St Thomas
College *Palai*

AUTONOMOUS

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ST.THOMAS COLLEGE PALAI (AUTONOMOUS)

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CERTIFICATE

This is to certify that the project entitled '**Impact of RBI Monetary Policy on Sectoral Stock Returns in India**' is a bonafide record of research work carried out by **ABHISHEK S P** Reg. No. **24PSTAT0031**, under my supervision and guidance, in partial fulfillment of the requirements for the award of the Degree of **Master of Science in Statistics**.

It is further certified that this work has not previously formed the basis for the award of any degree, diploma, fellowship, or any other similar title of any university or institution.

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DECLARATION

I hereby declare that the project entitled “**Impact of RBI Monetary Policy on Sectoral Stock Returns in India**” is a bonafide record of research work done by me under the supervision and guidance of **Ms. Reeba Mary Alex**, Department of Statistics, St Thomas College (Autonomous), Palai.

This project is submitted in partial fulfilment of the requirements for the award of the Degree of Master of Science in Statistics at Mahatma Gandhi University.

I further declare that this work has not previously formed the basis for the award of any degree, diploma, fellowship, or any other similar title of any university or institution. All sources used in this work have been duly acknowledged.

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Chapter 1

Introduction

1.1 Background of the Study

At regular intervals, monetary policy decisions in India are taken by the Monetary Policy Committee (MPC) of the Reserve Bank of India (RBI), which determines the repo rate — the rate at which the central bank lends to commercial banks, see [Reserve Bank of India \(2026\)](#). Changes in this policy rate, even when small, can influence financial markets, affecting borrowing costs, investment decisions, and overall economic activity.

The RBI, established in 1935, serves as the central monetary authority of the country. Its primary objective is to maintain price stability while supporting economic growth, with the repo rate acting as a key policy instrument, see [Reserve Bank of India \(2026\)](#). Although monetary transmission is theoretically well understood, its actual impact across financial markets tends to be uneven and complex.

Over the past two decades, India’s equity market has expanded significantly. The National Stock Exchange (NSE), particularly through indices such as the NIFTY 50, has seen increasing participation from both retail and institutional investors, see [National Stock Exchange \(2026\)](#). As a result, financial markets have become more responsive to macroeconomic signals, including monetary policy announcements.

However, the stock market does not respond uniformly. Sectoral differences play an important role in shaping market reactions. For instance, banking stocks are highly sensitive to interest rate changes, whereas sectors like information technology are influenced more by global demand conditions. Similarly, real estate depends on credit availability, while consumer-oriented sectors reflect spending behavior, see [Reserve Bank of India \(2019\)](#). These variations necessitate a disaggregated approach rather than treating the market as a single entity.

1.2 Statement of the Problem

Existing literature has extensively examined the relationship between monetary policy and stock market performance. For example, [Bernanke and Kuttner \(2005\)](#) demonstrate that unexpected policy rate changes significantly affect equity markets in the United States. In the Indian context, studies generally confirm that RBI policy decisions influence broad market indices such as the NIFTY 50 and Sensex, see [National Stock Exchange \(2026\)](#).

Despite these contributions, two major gaps remain.

First, most studies adopt an aggregate market perspective, overlooking sector-specific differences in response to policy changes. This aggregation may conceal important variations arising from differing sensitivities to interest rates and macroeconomic conditions.

Second, limited attention has been given to the timing of market reactions. It remains unclear whether stock prices adjust prior to policy announcements, on the announcement day, or afterward. Understanding this timing is crucial for evaluating market efficiency and information dissemination.

This study addresses these gaps by analyzing sectoral responses within a short event window around RBI monetary policy announcements.

1.3 Objectives of the Study

The following objectives guided this study:

1. To analyze the impact of RBI monetary policy announcements on short-term stock returns across different sectors using Abnormal Returns (AR) and Cumulative Abnormal Returns (CAR).
2. To estimate and compare sectoral sensitivity to repo rate changes using panel data regression techniques.
3. To examine the timing of market reactions by distinguishing between pre-announcement, announcement-day, and post-announcement effects.

1.4 Scope of the Study

This study covers the period from January 2011 to January 2026, encompassing multiple monetary policy phases, including tightening cycles, easing periods, and the COVID-19 pandemic.

The analysis focuses on six sectoral indices: NIFTY Bank, NIFTY IT, NIFTY Auto, NIFTY FMCG, NIFTY Pharma, and NIFTY Realty. These sectors were selected based on their differing sensitivity to interest rate movements and the availability of consistent data.

Stock market data were obtained from the National Stock Exchange (NSE), while policy decisions and announcement dates were sourced from official RBI publications. The analysis was conducted in R using packages such as `dplyr`, `plm`, `ggplot2`, `lubridate`, and `zoo`.

1.5 Significance of the Study

This study contributes to the literature by examining monetary policy transmission from a sectoral perspective in an emerging market context. By integrating event study methodology, panel regression, and timing analysis, it provides a more comprehensive understanding of market responses to policy changes.

The findings have practical implications for investors and portfolio managers in optimizing sector allocation around policy announcements. Additionally, they offer insights for policymakers regarding the effectiveness of policy transmission. Early market reactions may indicate information leakage, while heterogeneous sectoral responses could reflect structural differences within the economy.

Chapter 2

Literature Review

2.1 Theoretical Framework

This study builds on two core theories. The first is the Efficient Market Hypothesis (EMH), put forward by [Fama \(1970\)](#) which says that markets absorb all available information into prices almost immediately. Under its semi-strong form, only surprising information should cause abnormal returns — anything the market already expected should already be priced in. For monetary policy research, this means that if a rate change is widely anticipated, markets should barely react. It is the unexpected decisions that tend to move prices.

That said, whether EMH actually holds in emerging markets like India is still a matter of debate. Information asymmetry, behavioral biases, and uneven market efficiency all get in the way, raising genuine questions about how closely Indian markets follow the theory.

The second theory is the monetary policy transmission mechanism, which explains how central bank decisions work their way through the financial system and eventually reach the real economy. The equity market is one important channel here — interest rate changes affect discount rates, corporate earnings expectations, and investor risk appetite, all of which feed into stock prices. But how quickly and how strongly this happens differs a lot across sectors and markets, which is why looking only at the aggregate market can miss a lot.

2.2 International Evidence

One of the most cited works in this area is [Bernanke and Kuttner \(2005\)](#), who studied how US equity markets responded to Federal Reserve decisions. Their key finding was that markets mainly react to the surprise element of rate changes — expected decisions

barely move the needle, which lines up with EMH. They found that an unexpected 25 basis point cut is associated with roughly a one percent rise in broad equity indices.

[Rigobon and Sack \(2004\)](#) added an important layer by showing that while monetary policy does affect asset prices, the size and timing of the reaction can vary quite a bit, and short-term effects are not always statistically meaningful. This is a reminder that monetary transmission is complicated, and drawing broad conclusions from aggregate market data can be misleading.

[Flannery and Protopapadakis \(2002\)](#) showed that macroeconomic announcements do not hit all sectors equally — interest-sensitive sectors tend to react more sharply than defensive ones. On the methodology side, [Brown and Warner \(1985\)](#) laid the groundwork for event study analysis, which continues to be the standard approach for measuring abnormal returns around scheduled announcements.

While this international evidence is strong, most of it comes from developed markets, which tend to be more efficient than emerging ones.

2.3 Indian Studies

Research on this topic from India is relatively sparse, and the findings are mixed. [Pandey \(2005\)](#) found that RBI policy announcements often have weak short-term effects on Indian equity markets, with actual reactions frequently diverging from what theory would predict.

[Bhanumurthy et al. \(2014\)](#) reached similar conclusions — short-term transmission effects tend to be limited and inconsistent, and they vary depending on the type of policy change and the broader economic environment at the time.

These patterns reflect some real structural features of Indian financial markets: the heavy presence of public sector banks, a large retail investor base, and a market that is still developing in terms of efficiency. All of this means Indian markets may not always behave the way EMH-based models expect, which makes a more detailed and India-specific analysis all the more necessary.

2.4 Sectoral Studies

There is strong support in the literature for looking at sectors individually rather than as a whole. [Flannery and Protopapadakis \(2002\)](#) demonstrated that sectors respond differently to macroeconomic announcements — real estate and banking, being more sensitive to borrowing costs, tend to show stronger and more consistent reactions than

sectors with weaker links to interest rates.

This makes intuitive sense. A rate hike directly raises borrowing costs and squeezes credit availability, hitting sectors like real estate hard. For something like pharmaceuticals, the impact is far more indirect and muted.

Despite this, most Indian studies still rely on aggregate indices. The few sector-level studies that do exist tend to cover short time periods or a limited set of policy events, making it difficult to capture how monetary policy transmission plays out across different phases of the economic cycle.

2.5 Research Gap

The existing body of research makes a few things clear — monetary policy announcements do generate short-term market reactions, the surprise component of those announcements matters most, and sectoral differences are real and significant. But there are still notable gaps, especially in the Indian context.

Most studies either rely on aggregate market indices, ignore the timing dimension of reactions, or use just one methodological approach. This limits how fully we can understand the relationship between monetary policy and different sectors over time.

This study tries to fill those gaps by analyzing fifteen years of RBI policy decisions across six sectoral indices, combining event study methodology, panel regression, and timing analysis within a single framework. The goal is to get a more complete and nuanced picture of how monetary policy shocks actually affect sectoral equity returns in India.

Chapter 3

Data and Methodology

3.1 Data Description

This study uses daily closing price data for six sectoral indices — NIFTY Bank, NIFTY IT, NIFTY Auto, NIFTY FMCG, NIFTY Pharma, and NIFTY Realty — with NIFTY 50 serving as the broad market benchmark. The data runs from January 2011 to January 2026, covering roughly fourteen years of trading activity. All index data were taken from the National Stock Exchange (NSE) of India, see [National Stock Exchange \(2026\)](#).

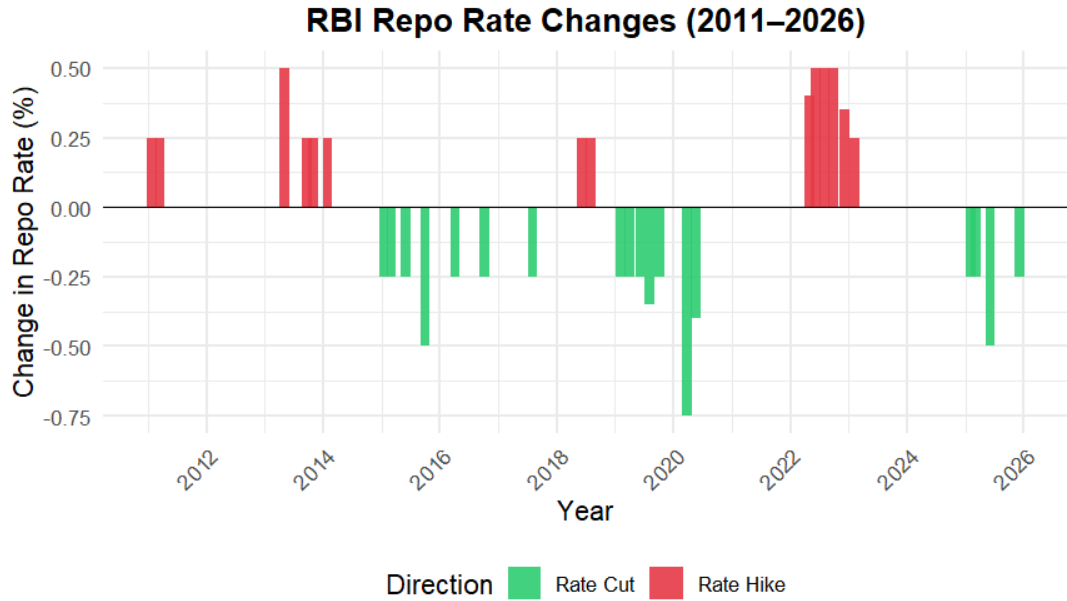
Daily logarithmic returns were computed for each index as:

$$R_{it} = \ln \left(\frac{P_{it}}{P_{it-1}} \right) \quad (3.1)$$

where P_{it} is the closing price of sector i on day t . Log returns were chosen over simple returns because of their better statistical properties — they are symmetric and time-additive, which makes them more suitable for this kind of analysis, see [Campbell \(1997\)](#).

RBI monetary policy announcement dates and the corresponding repo rate decisions were collected directly from the RBI's official website, see [Reserve Bank of India \(2026\)](#). Only dates on which the Monetary Policy Committee made a formal policy decision were treated as events. Trading holidays with missing data were excluded from the event window. After cleaning the data, **61 policy events** were identified across the study period.

All analysis was performed in R, using packages including `dplyr`, `zoo`, `lubridate`, `plm`, and `ggplot2`.



Source: Reserve Bank of India

Figure 3.1: Policy Change Bar Chart

3.2 Event Study Methodology

The event study approach, developed by [Brown and Warner \(1985\)](#) and further formalized by [MacKinlay \(1997\)](#), is a widely used method for measuring how specific events affect asset prices. The core idea is simple — if an event brings new information to the market, it should cause returns to deviate from what would normally be expected. These deviations are called **Abnormal Returns (AR)**.

Defining the Event Window

For each RBI policy announcement, an event window of seven trading days was used, starting three days before the announcement and ending three days after: $[t - 3, t + 3]$, with the announcement date itself at the center. This window is wide enough to pick up both early market reactions and any delayed responses, while still being narrow enough to avoid noise from unrelated events.

Estimation Window

To estimate expected returns, an estimation window of 120 trading days prior to the event window was used: $[t - 123, t - 4]$. This period is used to estimate the parameters of the market model for each sector.

The Market Model

Expected returns for each sector are estimated using the market model [MacKinlay \(1997\)](#):

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \quad (3.2)$$

where R_{it} is the return of sector i on day t , R_{mt} is the NIFTY 50 market return on day t , α_i and β_i are sector-specific parameters estimated over the estimation window, and ε_{it} is the error term.

Abnormal and Cumulative Abnormal Returns

The abnormal return for sector i on event day τ is the difference between the actual return and what the market model predicted:

$$AR_{i\tau} = R_{i\tau} - (\hat{\alpha}_i + \hat{\beta}_i R_{m\tau}) \quad (3.3)$$

Cumulative Abnormal Returns (CAR) are then calculated by adding up these abnormal returns across the full event window:

$$CAR_i = \sum_{\tau=-3}^{+3} AR_{i\tau} \quad (3.4)$$

CAR gives a picture of the total market reaction over the entire event window, rather than focusing on just one day.

Significance Testing

A cross-sectional t-test was used to check whether the abnormal returns were statistically meaningful. The null hypothesis is that the average abnormal return across all events is zero, meaning the announcement had no real impact. Rejection at the 5% significance level confirms a statistically significant market reaction.

3.3 Panel Data Regression

The event study captures short-term reactions around announcement dates, but it does not say much about the overall sensitivity of each sector to monetary policy across the full study period. Panel data regression was used to fill that gap, see [Baltagi \(2008\)](#).

Data Structure

The panel dataset includes six sectors observed across the entire study period, forming a balanced panel. The dependent variable is the daily sectoral return R_{it} , and the key independent variable is the change in the repo rate $\Delta Repo_t$, which takes a non-zero value

only on policy announcement dates and is zero on all other days. The NIFTY 50 return R_{mt} is included as a control variable to account for general market movements.

Model Specification

The baseline panel regression model is:

$$R_{it} = \alpha_i + \beta_1 \Delta Repo_t + \beta_2 R_{mt} + \varepsilon_{it} \quad (3.5)$$

where α_i represents sector-specific effects, β_1 measures how sensitive sectoral returns are to repo rate changes, and β_2 captures the relationship with overall market returns.

Fixed Effects vs Random Effects

Choosing between Fixed Effects and Random Effects models is an important decision in panel regression. The Fixed Effects model controls for all time-invariant characteristics that differ across sectors, while the Random Effects model assumes those characteristics are unrelated to the explanatory variables.

The **Hausman test** [Hausman \(1978\)](#) was used to decide which model fits better. If the null hypothesis — that Random Effects is consistent and efficient — is rejected, Fixed Effects is preferred. The appropriate model was selected based on these test results.

To further examine sectoral differences, the model was also estimated separately for each sector, allowing β_1 to vary and helping identify which sectors are more responsive to repo rate changes.

3.4 Timing Analysis

The third objective focuses on identifying the timing of market reactions — specifically, when sectoral responses occur relative to the policy announcement.

To do this, the average abnormal return for each event day $\tau \in \{-3, -2, -1, 0, +1, +2, +3\}$ was calculated separately for each sector and tested for statistical significance. This day-by-day breakdown helps identify three possible patterns. Significant abnormal returns on days $t - 1$ or $t - 2$ suggest the market was already reacting ahead of the announcement. Significant returns on $t = 0$ point to an immediate reaction, consistent with efficient information processing. Significant returns on $t + 1$ or later suggest the market was slow to absorb the information, which may indicate some degree of inefficiency.

Comparing these patterns across sectors gives a clearer sense of how efficiently different parts of the Indian equity market respond to monetary policy signals.

Chapter 4

Results and Discussion

4.1 Exploratory Data Analysis

The six sectoral return series span fifteen years and thousands of trading days. Before getting into formal analysis, it is worth stepping back and asking — what does this data actually look like? The following section examines the distributional properties, stationarity, and relationships between sectoral returns, and in doing so, it already begins to tell the story that the later analysis confirms.

4.1.1 Descriptive Statistics

Table 4.1 presents the summary statistics of daily logarithmic returns for the six sectoral indices over the study period from January 2011 to January 2026.

Sector	Mean	Std Dev	Min	Max	Skewness	Kurtosis
NIFTY Bank	0.000392	0.014456	-0.1831	0.1000	-0.6184	14.5201
NIFTY IT	0.000439	0.013330	-0.1249	0.0892	-0.5194	10.3891
NIFTY Auto	0.000514	0.013329	-0.1491	0.0990	-0.4056	11.3859
NIFTY FMCG	0.000468	0.010388	-0.1120	0.0799	-0.3182	11.4815
NIFTY Pharma	0.000400	0.011546	-0.0935	0.0986	-0.1826	7.9741
NIFTY Realty	0.000227	0.019989	-0.1233	0.0830	-0.4429	5.7484

Table 4.1: Descriptive Statistics of Daily Sectoral Returns

The first thing that stands out is that all six sectors delivered positive average daily returns over this period, which is consistent with the long-term upward trend in Indian equities. But the differences between sectors are telling.

NIFTY Auto had the highest average daily return at 0.000514, while NIFTY Realty had

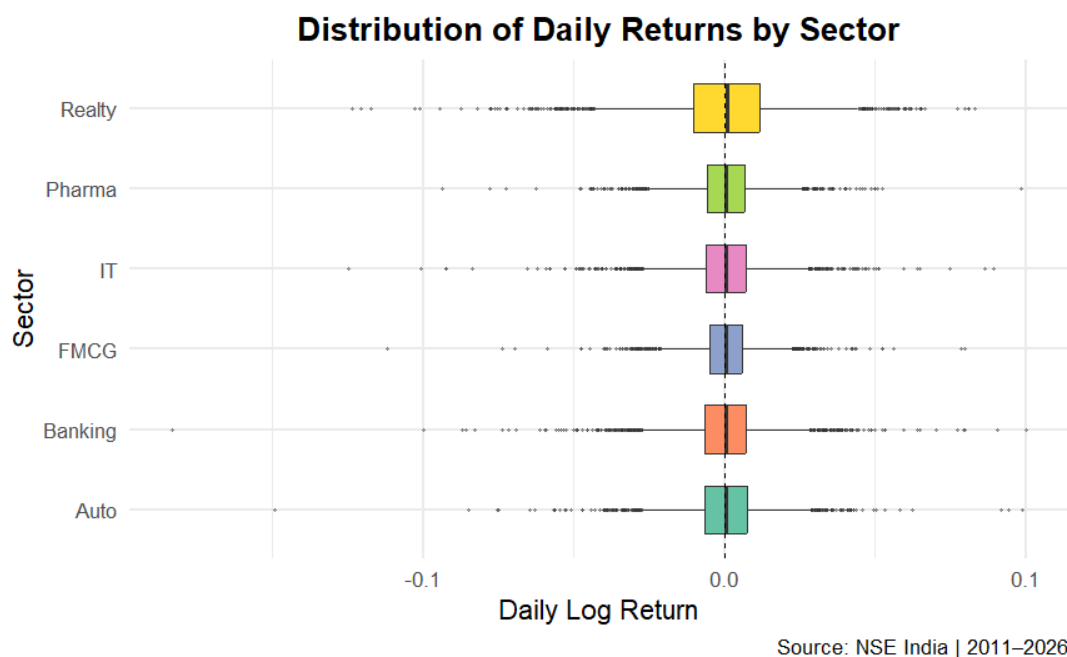


Figure 4.1: Boxplot for Distribution of Daily Returns

the lowest at 0.000227. That gap might seem small on a daily basis, but compounded over fifteen years it represents a meaningful divergence. What makes Realty’s position even more interesting is that it also carried the highest risk — a standard deviation of 0.019989, nearly double that of NIFTY FMCG at 0.010388, which was the most stable sector. In other words, Realty investors absorbed the most volatility for the least return. This is not a coincidence — it reflects how deeply the sector is tied to interest rate cycles. When rates rise, property demand softens, home loan EMIs climb, and developers struggle to service their debt. The sector pays a price that most others simply do not.

Every sector also shows negative skewness, meaning large losses happen more often than comparably large gains. This is a well-known feature of equity markets — investors tend to sell faster than they buy in a panic. The kurtosis values reinforce this. All six sectors have kurtosis well above 3, confirming that extreme daily moves — both sharp crashes and sudden rallies — occur far more often than a normal distribution would suggest. NIFTY Bank sits at the extreme end with a kurtosis of 14.52, which reflects just how violently the banking sector can move when something unexpected happens — a rate surprise, a banking sector crisis, or a shift in global liquidity. Realty, with a kurtosis of 5.75, is comparatively calmer at the extremes, even if its day-to-day swings are the largest of the group.

4.1.2 Normality Test

Knowing the data looked non-normal was one thing. Confirming it formally was another. The Jarque-Bera test, see [Jarque and Bera \(1980\)](#) was applied to each sector's return series to test whether the distribution is normal. The results were clear — every sector rejected normality at the 1% significance level, with p-values effectively at zero. The test statistics were enormous, ranging from 1,291.39 for Realty to 20,992.28 for Banking. These are not borderline cases. The data is clearly non-normal, and any method that assumes otherwise would be on shaky ground. This finding directly justifies the use of robust testing procedures in the event study that follows.

4.1.3 Stationarity Test

Working with non-stationary data in regression analysis is problematic — it can produce results that look meaningful but are entirely spurious. The Augmented Dickey-Fuller (ADF) test, see [Dickey and Fuller \(1979\)](#) was therefore applied to each return series to check for unit roots.

Sector	ADF Statistic	Stationary
NIFTY Bank	-15.3255	Yes
NIFTY IT	-15.1210	Yes
NIFTY Auto	-14.8423	Yes
NIFTY FMCG	-16.4661	Yes
NIFTY Pharma	-16.2255	Yes
NIFTY Realty	-14.5139	Yes

Table 4.2: ADF Test Results for Sectoral Return Series

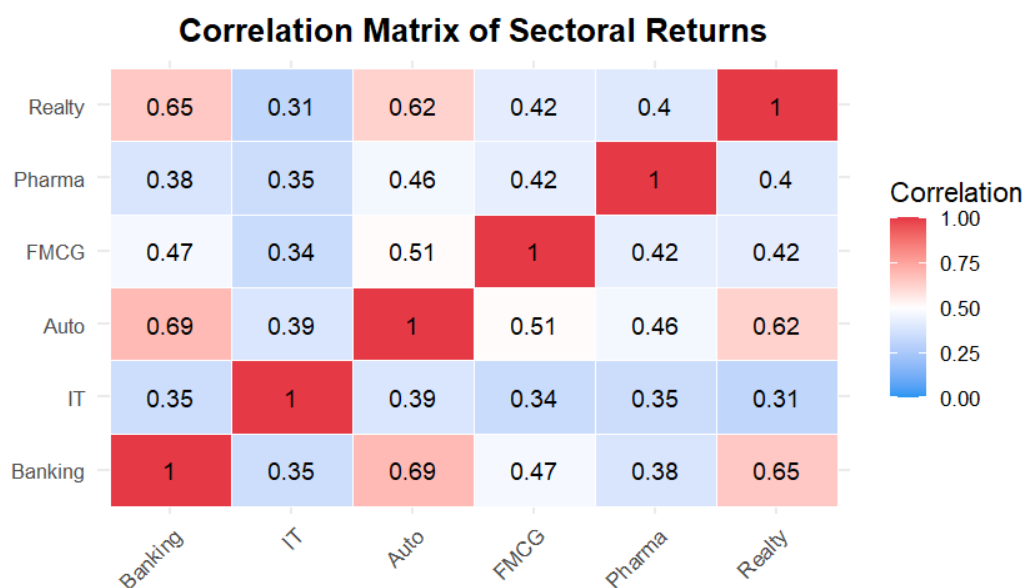
All six series are stationary at the 1% level, with ADF statistics ranging from -14.51 for Realty to -16.47 for FMCG. This is a clean and expected result — logarithmic returns, unlike raw price levels, do not drift over time and do not carry unit roots. The analysis can proceed on solid statistical footing.

4.1.4 Correlation Analysis

The correlation matrix is where the data starts to group itself into a story. Three sectors — Banking, Auto, and Realty — form a noticeably tight cluster. Banking and Auto move together with a correlation of 0.688. Banking and Realty sit at 0.650. Auto and Realty at 0.622. These are not mild associations — they reflect a shared economic logic. All three sectors are fundamentally credit-driven. When interest rates move, these three feel it first and hardest. Car loans get more expensive, home loans get more expensive,

	Banking	IT	Auto	FMCG	Pharma	Realty
Banking	1.000	0.349	0.688	0.471	0.380	0.650
IT	0.349	1.000	0.388	0.341	0.345	0.311
Auto	0.688	0.388	1.000	0.510	0.463	0.622
FMCG	0.471	0.341	0.510	1.000	0.425	0.416
Pharma	0.380	0.345	0.463	0.425	1.000	0.401
Realty	0.650	0.311	0.622	0.416	0.401	1.000

Table 4.3: Correlation Matrix of Sectoral Daily Returns



Source: NSE India | 2011–2026

Figure 4.2: Correlation Heatmap

and bank margins widen or narrow depending on the direction of the move. It makes complete sense that they tend to move together.

IT sits apart from the rest. Its highest correlation is 0.388 with Auto, and with Realty it is just 0.311 — the lowest pair in the entire matrix. This is exactly what one would expect. Indian IT companies earn predominantly in dollars, serve foreign clients, and are largely insulated from domestic monetary policy. The RBI repo rate simply does not feature much in their world.

Pharma sits in the middle ground — moderate correlations with every sector, no strong pull toward any cluster. Like IT, it is a defensive sector, but unlike IT, it has more domestic exposure through hospital supply chains and government procurement, which keeps it modestly connected to the broader market.

This grouping — interest-rate-sensitive sectors on one side, defensives on the other — is not just an interesting observation. It directly predicts what the event study and panel regression will find. Sectors that share the same economic exposures tend to share the same reactions, and monetary policy is one of the most powerful shared exposures in the Indian market.

4.2 Event Study Results

4.2.1 Market Model Estimates

Before abnormal returns can be computed, the market model needs to be estimated for each sector over the 120-day estimation window. Table 4.4 presents the sector-wise alpha and beta estimates from these regressions.

Sector	Alpha	Beta
NIFTY Bank	-0.000025	1.2166
NIFTY IT	0.000126	0.7565
NIFTY Auto	0.000129	1.0308
NIFTY FMCG	0.000254	0.6354
NIFTY Pharma	0.000136	0.5978
NIFTY Realty	-0.000147	1.3229

Table 4.4: Market Model Estimates by Sector

The beta coefficients reveal clear sectoral differences. Realty has the highest beta at 1.32, meaning it amplifies broad market movements by roughly 32%. Banking follows closely at 1.22, while Auto is nearly aligned with the market at 1.03. These three can be described as the more market-sensitive sectors.

FMCG (0.64) and Pharma (0.60), on the other hand, behave more defensively — they absorb only a portion of overall market swings. This is consistent with the relatively stable demand conditions in these industries. IT, with a beta of 0.76, falls somewhere between the two groups, partly because of its foreign revenue exposure which offers some insulation from domestic market moves.

4.2.2 Cumulative Abnormal Returns

Table 4.5 presents the mean Cumulative Abnormal Returns (CAR) for each sector across the full event window, along with results broken down by rate hike and rate cut events.

The results tell a clear story. Among the six sectors, only NIFTY Realty shows statistically significant cumulative abnormal returns around RBI policy announcements. Its overall p-value of 0.0045 is well below the 1% threshold, driven mainly by rate hike

Sector	CAR (Hike)	CAR (Cut)	p-value (Overall)	p-value (Hike)	p-value (Cut)
NIFTY Bank	-0.00039	-0.00655	0.3284	0.9574	0.1004
NIFTY IT	-0.00300	+0.00616	0.5869	0.6721	0.0953
NIFTY Auto	-0.00503	-0.00344	0.2864	0.2221	0.5932
NIFTY FMCG	-0.00072	-0.00422	0.6103	0.9323	0.5322
NIFTY Pharma	+0.00456	-0.00442	0.9477	0.4973	0.6115
NIFTY Realty	-0.02542	-0.01800	0.0045	0.0071	0.1209

Table 4.5: Mean CAR by Sector and Policy Direction

events where the mean CAR reached -2.54%. All other sectors fail to show statistically significant abnormal returns within the event window.

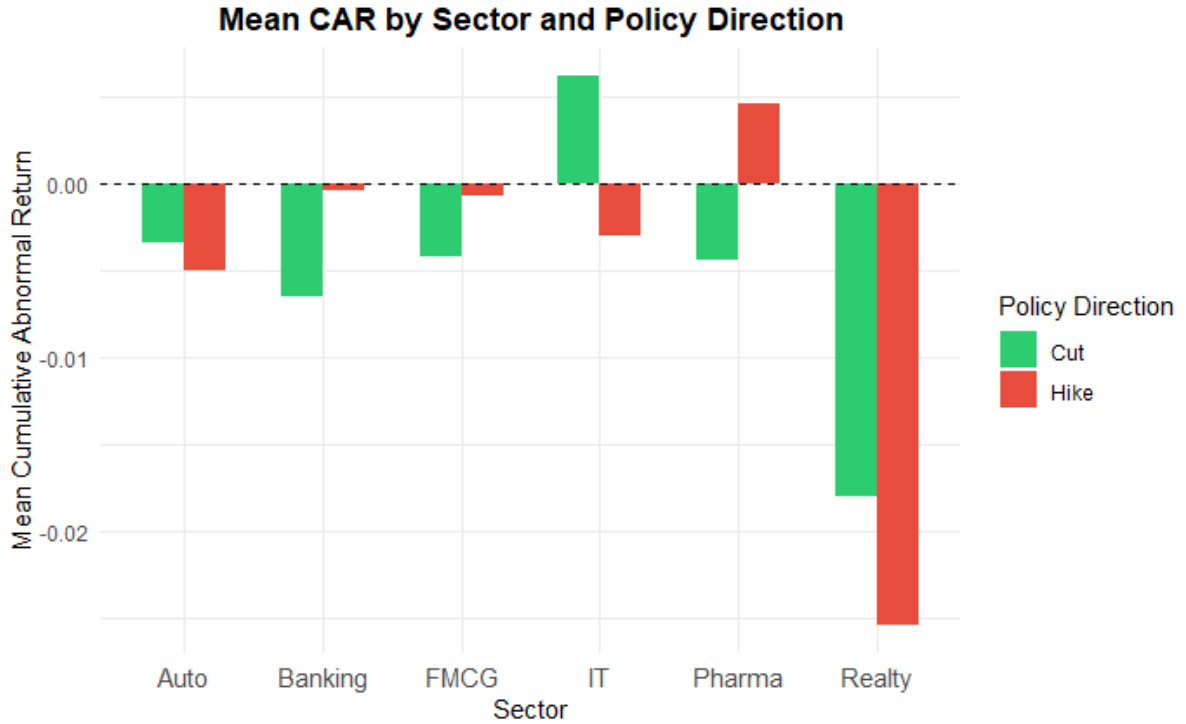


Figure 4.3: Mean Cumulative Abnormal Returns by Sector and Policy Direction

This result makes economic sense. Realty is among the most interest-rate-sensitive sectors — higher rates push up borrowing costs, weaken housing demand, and put pressure on developers who are already carrying significant debt. A CAR of around -2.5% during rate hike events reflects a real and meaningful market response.

The reaction during rate cuts is less clear-cut. The CAR remains negative but is not statistically significant. One possible explanation is that rate cuts often happen against the backdrop of broader economic slowdowns, where negative sentiment in the market may cancel out the benefit of cheaper credit.

IT shows a directionally sensible pattern — positive CAR during rate cuts and negative during hikes. But these effects are not statistically significant, suggesting the relationship

exists but is not strong enough to be conclusive within this sample.

4.2.3 Day-Zero Reaction

An important question is whether markets react on the announcement day itself. The results show that no sector displays statistically significant abnormal returns on day $t = 0$. Even the lowest p-value remains above conventional significance thresholds.

This suggests that market reactions to RBI announcements are not concentrated in a single day but are instead spread across the event window. This finding makes the timing analysis that follows all the more relevant.

4.2.4 Day-Wise Abnormal Returns

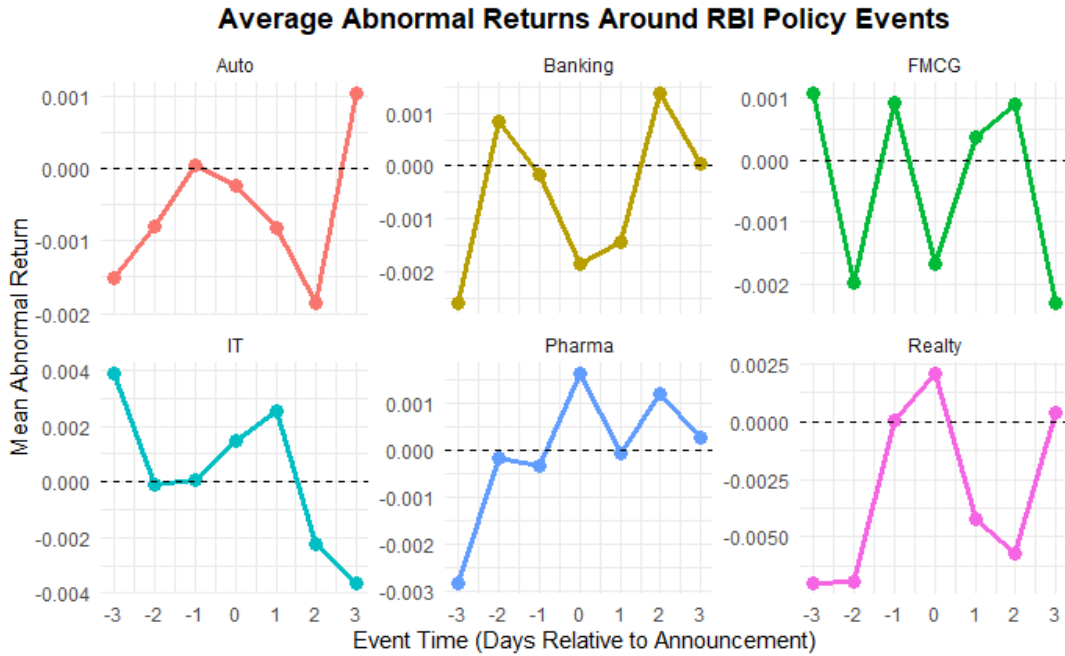


Figure 4.4: Mean Abnormal Returns by Event Day and Sector ($t = -3$ to $t = +3$)

The day-wise analysis adds useful detail to the timing of market responses. Realty shows a noticeable negative movement around $t = -1$ and $t = 0$, suggesting that part of the market reaction may already be happening before the official announcement — possibly reflecting anticipatory behavior or pre-formed market expectations.

IT shows a reversal pattern, with positive movements before the announcement and a correction in the post-event period, pointing to delayed adjustment. Banking fluctuates around zero throughout, which is consistent with its overall lack of statistical significance.

These patterns reinforce the broader finding that monetary policy effects are not the same across sectors and tend to unfold gradually rather than all at once.

Overall, the event study results suggest that RBI monetary policy announcements produce limited short-term abnormal returns across most sectors of the Indian equity market. The clear exception is NIFTY Realty, which shows a consistent and statistically significant response. This aligns with the theoretical sensitivity of real estate to interest rate movements and is in line with prior evidence on interest-rate-sensitive sectors.

4.3 Panel Data Regression Results

4.3.1 Overall Panel Model

The Hausman test points to the Fixed Effects model as the more appropriate choice, suggesting that each sector carries its own underlying characteristics that need to be accounted for. The panel dataset covers six sectors over approximately 3,700 trading days each, giving a total of 22,123 observations. The results are presented in Table 4.6.

Variable	Coefficient	Std. Error	t-value	p-value
Repo Rate Change	0.00032	0.00528	0.061	0.9515
Market Return	0.92480	0.00669	138.279	< 0.001***
Change $\times$ Bank	-0.00155	0.00747	-0.208	0.8354
Change $\times$ FMCG	-0.00030	0.00747	-0.041	0.9675
Change $\times$ IT	-0.00054	0.00747	-0.073	0.9421
Change $\times$ Pharma	-0.00150	0.00747	-0.201	0.8409
Change $\times$ Realty	-0.00772	0.00747	-1.035	0.3009

Table 4.6: Fixed Effects Panel Regression Results

R-Squared: 0.4638 Adjusted R-Squared: 0.4635 F-statistic: 2731.88 ($p < 0.001$)

The dominant driver in this model is the market return. With a coefficient of 0.9248, significant well below the 1% level, sectoral returns move very closely with the broader market on a typical trading day. This is expected — sectoral indices are inherently tied to overall market conditions and sentiment.

What is more revealing is the role of the repo rate. Its coefficient is essentially zero (0.00032), with a p-value of 0.9515, indicating no statistically significant relationship with sectoral returns in the panel framework. None of the interaction terms between repo rate changes and individual sectors are significant either. Even NIFTY Realty — which showed clear responses in the event study — produces a coefficient of -0.00772 with a p-value of 0.30. While the direction aligns with expectations, the effect is not statistically distinguishable from zero when averaged across all trading days.

4.3.2 Sector-wise Sensitivity

Sector	Beta (Repo Rate)	Direction
Realty	-0.007404	Negative
Banking	-0.001230	Negative
Pharma	-0.001178	Negative
Auto	+0.000321	Positive
IT	-0.000221	Negative
FMCG	+0.000016	Positive

Table 4.7: Sector-wise Repo Rate Sensitivity from Panel Regression

Figure 4.5 illustrates the relative size of these sensitivities. Realty stands out clearly, with a coefficient of -0.0074 — substantially larger in magnitude than any other sector. Banking comes next at a much lower level, while the remaining sectors show only minor differences.

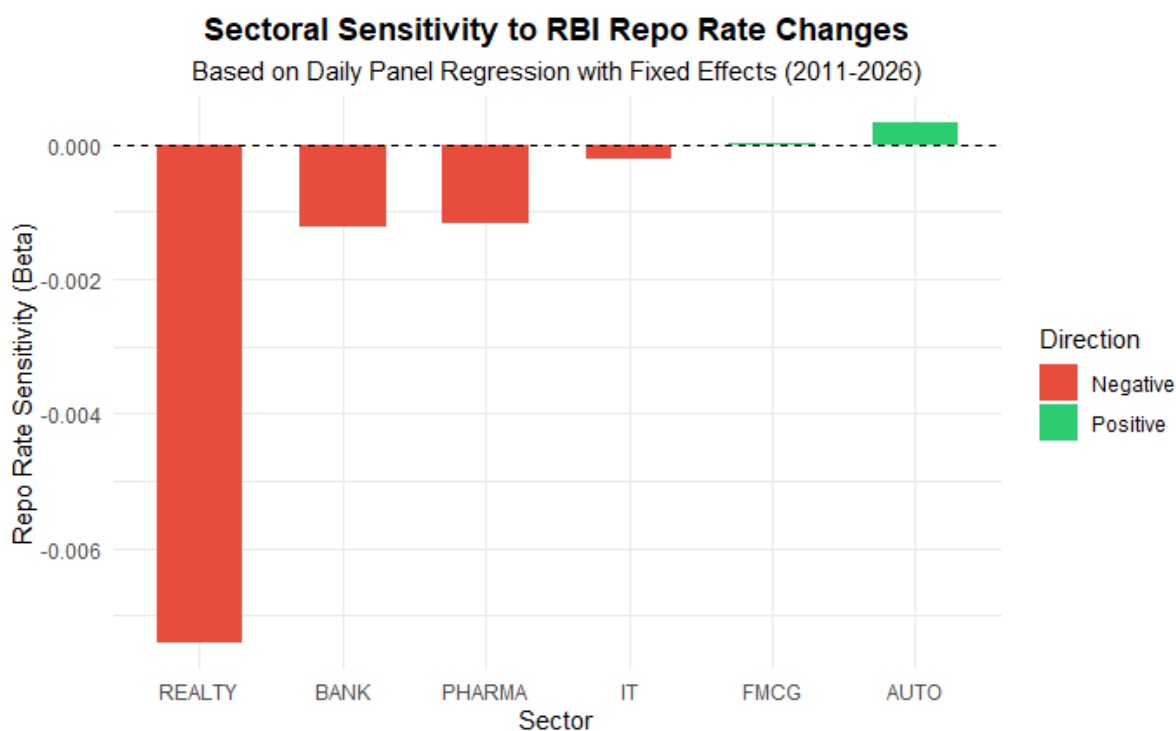


Figure 4.5: Sector-wise Repo Rate Sensitivity — Fixed Effects Panel Regression

The direction of these coefficients broadly makes economic sense. Realty and Banking are both heavily dependent on credit conditions. When interest rates rise, borrowing becomes more expensive, housing demand tends to soften, and lending activity may slow — all of which point toward negative sensitivities.

Pharma shows a smaller negative coefficient, which likely reflects indirect co-movement with broader market conditions rather than a direct link to monetary policy. Auto and FMCG show small positive coefficients, which may initially seem counterintuitive but is

plausible in the Indian context — rate cuts often coincide with broader economic stimulus and improving consumer confidence.

FMCG’s coefficient is almost exactly zero, suggesting it is largely unaffected by repo rate changes. This is consistent with its reliance on stable, consumption-driven demand rather than credit cycles.

4.3.3 Why the Panel Results Differ from the Event Study

At first glance, the panel regression results might seem to contradict what the event study found. The event study showed a statistically significant response for NIFTY Realty, while the panel regression found no significant effects at all. But these two results are not actually in conflict — they are just measuring different things.

The key difference comes down to how the data is structured in each approach. The event study zooms in on a tight seven-day window around each policy announcement, where the effect of monetary policy is most visible and least cluttered by other market noise. The panel regression, on the other hand, covers every single trading day over the full fourteen-year period. On most of those days, there is no policy change, so the repo rate variable simply sits at zero. When you average the effect across thousands of such days, the impact naturally gets diluted — not because monetary policy does not matter, but because its effects are concentrated around specific events rather than spread evenly across time.

So rather than one result undermining the other, the two approaches together tell a more complete story. Monetary policy does not seem to move sectoral returns on an ordinary trading day. But around policy announcement dates, certain sectors — Realty in particular — show clearer and more consistent reactions.

4.4 Timing Analysis

The event study established that Realty is the only sector generating statistically significant cumulative abnormal returns. However, CAR is a summary measure that compresses movements across multiple days into a single figure. The timing analysis therefore examines a more specific question: on which days do sectors actually react, and what does that reveal about how the market is processing monetary policy information?

Table 4.8 presents the aggregate day-wise abnormal returns across all sectors, while Table 4.9 reports only the statistically significant sector-level responses.

At the aggregate level, no event day crosses the 5% significance threshold. The closest are $t = -3$ and $t = -2$, both just above conventional levels, suggesting a mild negative drift

Event Day	Mean AR	t-statistic	p-value
$t = -3$	-0.00149	-1.94	0.0537
$t = -2$	-0.00152	-1.82	0.0710
$t = -1$	+0.00009	0.128	0.8980
$t = 0$	+0.00025	0.341	0.7340
$t = +1$	-0.00061	-0.784	0.4340
$t = +2$	-0.00106	-1.38	0.1700
$t = +3$	-0.00070	-1.09	0.2780

Table 4.8: Aggregate Day-wise Abnormal Returns Around RBI Announcements

before announcements. The announcement day itself ($t = 0$) shows almost no reaction at all, with a near-zero mean abnormal return and a high p-value. This again supports the finding that market responses are not concentrated on any single day but are instead dispersed across time and across sectors.

Sector	Event Day	Mean AR	p-value
IT	$t = -3$	+0.00391	0.0413
IT	$t = +3$	-0.00366	0.0313
Realty	$t = -3$	-0.00702	0.0207
Realty	$t = -2$	-0.00691	0.0451
Realty	$t = +2$	-0.00570	0.0085

Table 4.9: Significant Day-wise Abnormal Returns by Sector ($p < 0.05$)

Two sectors stand out — Realty and IT — and they display very different timing patterns.

Realty — The Anticipatory Sector

Realty shows statistically significant negative abnormal returns at $t = -3$, $t = -2$, and $t = +2$, but not on the announcement day itself. This tells us that a meaningful part of the adjustment happens before the policy decision is formally announced.

This pattern is consistent with anticipatory behavior. In the days leading up to an MPC meeting, market participants are exposed to a range of public signals — inflation data, central bank communication, and analyst forecasts. For a sector as sensitive to interest rates as Realty, these signals appear sufficient to trigger early repositioning.

The reappearance of a significant negative return at $t = +2$ suggests a second phase of adjustment, possibly reflecting a more detailed reading of the policy statement beyond the headline rate decision. Market reactions for Realty, in other words, seem to unfold in stages rather than all at once.

IT — The Reversal Sector

IT shows a contrasting pattern. A positive abnormal return appears at $t = -3$, followed by a negative return at $t = +3$. This suggests a reversal over the event window.

One possible explanation is that in the days before the announcement, uncertainty around currency movements may temporarily benefit IT, given its export-oriented nature. Once the announcement is made and the broader macroeconomic picture becomes clearer, this positioning gets unwound.

Another interpretation is that IT serves as a kind of defensive allocation during periods of policy uncertainty, attracting inflows ahead of the announcement, which are then reversed once the uncertainty clears.

Either way, the net effect across the full event window cancels out, which explains why IT's cumulative abnormal return is not statistically significant.

Implications for Market Efficiency

These timing patterns offer insight into how monetary policy information is being incorporated into prices. Under the semi-strong form of the Efficient Market Hypothesis, asset prices should reflect publicly available information without delay.

The evidence here suggests a more nuanced outcome. Realty's pre-announcement adjustment indicates that publicly available information is already being priced in before the formal decision. This does not necessarily contradict market efficiency — it reflects forward-looking behavior based on available signals.

At the same time, the absence of significant announcement-day reactions across all sectors suggests that policy decisions are largely anticipated by the time they are officially made. Market responses appear to be shaped more by expectations and subsequent interpretation than by the announcement itself.

Overall, monetary policy effects are not instantaneous or uniform — they vary across sectors and play out over time.

4.5 Integrated Discussion

Three methods, three lenses — one coherent story.

The event study, panel regression, and timing analysis are not meant to be read separately. Each method answers a different question, and it is only when they are put together that a clearer picture emerges of how RBI monetary policy moves through India's equity markets.

The central finding that runs through all three methods is consistent: Realty is the most sensitive sector to RBI monetary policy, while most other sectors do not show statistically significant short-run responses. This is not a result of one particular model or specification — it holds across different methodological approaches, across a long time period, and across both rate hike and rate cut cycles.

The event study captures the immediate impact of policy announcements. When the RBI announces a rate hike, Realty generates a cumulative abnormal return of -2.54% — the only sector to do so with statistical significance. The panel regression confirms the relative ordering — Realty has the largest repo rate sensitivity coefficient by a clear margin, even if it does not reach statistical significance when averaged across all trading days. The timing analysis adds another dimension by showing that Realty begins adjusting at $t = -3$ and $t = -2$, meaning the market is already incorporating available signals before the formal announcement.

What makes this finding particularly compelling is that it holds consistently across all three methods. Each approach has its own limitations — the event study depends on the choice of window, the panel regression dilutes the signal across many non-event days, and the timing analysis works with smaller subsamples. Yet despite these differences, Realty emerges as the most responsive sector in every case. That consistency strengthens the result considerably.

The remaining sectors tell an equally important story through their relative silence. Banking, despite its theoretical sensitivity to interest rates, does not produce statistically significant abnormal returns in the event study and shows only a modest, insignificant coefficient in the panel regression. This may reflect structural features of the Indian banking system — including the dominant role of public sector banks and the influence of factors like government policy, credit quality, and global capital flows — that dilute the direct effect of repo rate changes.

Auto and FMCG show similarly muted responses. These sectors are more closely tied to consumption patterns and broader economic conditions, which evolve over longer time horizons and are influenced by many factors beyond short-term rate changes. Pharma appears even more insulated, with near-zero sensitivity across all three methods.

IT presents a secondary finding worth noting. Its reversal pattern in the timing analysis — rising before announcements and correcting afterward — suggests it may function as a temporary allocation during periods of policy uncertainty rather than being directly driven by RBI decisions. The overall effect cancels out over the event window, which is why IT does not show up as significant in the cumulative abnormal return analysis. This is a good example of why the timing analysis adds value that aggregate measures miss.

At the broad level, the finding that RBI policy announcements generate limited abnormal returns is consistent with earlier work such as Pandey (2005) and Bhanumurthy et al. (2014), which also report weak short-run monetary policy transmission in India. It is also broadly in line with Bernanke and Kuttner (2005), who argue that market reactions are driven primarily by the unexpected component of policy changes. In a setting where policy decisions are increasingly signaled in advance through forward guidance and communication, the scope for genuine surprise — and therefore for strong immediate reactions — may simply be reduced.

What this study adds to the existing literature is the sectoral perspective. While the aggregate effect of monetary policy appears limited, this hides important variation across sectors. Realty absorbs a disproportionately large share of the policy impact, while most other sectors remain relatively unaffected in the short run.

For investors and analysts, this distinction matters. Treating the market as a single entity risks overlooking these differences, whereas a sector-focused approach gives a clearer and more actionable understanding of how monetary policy filters through different parts of the market.

Chapter 5

Summary, Conclusion and Recommendations

5.1 Summary of the Study

This study set out to examine how RBI monetary policy announcements affect sectoral stock returns in India — not at the level of the broad market, but sector by sector, day by day. Using fifteen years of daily data from January 2011 to January 2026, covering six sectoral indices and over sixty RBI policy events, three methodologies were used in sequence: an Event Study to capture abnormal returns around announcement dates, a Fixed Effects Panel Regression to estimate sector-wise sensitivity to repo rate changes, and a Timing Analysis to identify when each sector reacts within the event window.

The study was motivated by two gaps in the existing literature — the tendency to treat the Indian stock market as a single entity rather than a collection of structurally different sectors, and the limited attention given to the timing of market reactions. Both gaps are directly addressed through the findings.

5.2 Major Findings

The results across all three methods point to a consistent set of conclusions.

From the Event Study: Of the six sectors examined, only NIFTY Realty generated statistically significant cumulative abnormal returns around RBI policy announcements. Its overall p-value of 0.0045 was well below the 1% threshold, driven mainly by rate hike events where the mean CAR reached -2.54%. All other sectors — Banking, IT, Auto, FMCG, and Pharma — did not produce statistically significant abnormal returns. No sector showed a significant reaction on the announcement day itself, suggesting that

market responses are spread across the event window rather than concentrated at a single point.

From the Panel Regression: The Fixed Effects model shows that market returns account for a large portion of the day-to-day variation in sectoral returns, leaving limited room for monetary policy to explain additional variation on an average trading day. The repo rate change coefficient is statistically insignificant across all sectors. However, the ordering of sector-wise sensitivities remains economically meaningful — Realty carries the largest negative beta at -0.0074, followed by Banking at -0.0012, consistent with the expectation that credit-dependent sectors are more exposed to interest rate changes.

From the Timing Analysis: Realty’s significant reactions are concentrated at $t = -3$ and $t = -2$ — days before the formal announcement — and again at $t = +2$, pointing to a combination of anticipatory behavior and delayed adjustment. IT displays a reversal pattern, rising at $t = -3$ and correcting at $t = +3$, consistent with short-term positioning that unwinds after uncertainty clears. These patterns indicate a market that incorporates publicly available signals ahead of formal announcements, rather than reacting only when the decision is released, see [Fama \(1970\)](#).

5.3 Conclusion

The central conclusion of this study is that RBI monetary policy announcements generate limited and largely insignificant short-term abnormal returns across most sectors of the Indian equity market. But this aggregate picture hides an important sectoral difference. Realty stands out as the sector where monetary policy leaves a consistent and measurable impact across multiple methods and policy cycles.

This result reflects the underlying economic structure of the sector. Real estate is closely tied to interest rates through borrowing costs, financing conditions, and demand sensitivity. As a result, changes in the policy rate are more directly transmitted to Realty than to most other sectors. In contrast, other sectors are influenced by a broader range of factors that dilute the immediate effect of monetary policy — including institutional structures in banking, global revenue exposure in IT, and demand stability in FMCG and Pharma.

The relatively weak overall transmission observed in this study is consistent with earlier findings from India and suggests that monetary policy signals are not uniformly or immediately reflected in equity prices [Pandey \(2005\)](#); [Bhanumurthy et al. \(2014\)](#). Instead, their impact appears to be uneven across sectors and distributed over time.

5.4 Implications

For investors and portfolio managers, the results underline the importance of sector-level analysis. Realty is the sector most responsive to interest rate expectations. Positioning decisions around RBI announcements may therefore benefit more from focusing on sectoral exposure than on broad market movements. IT's pre-announcement behavior may also offer short-term opportunities, though its reversal pattern suggests that timing matters a lot.

For policymakers and the RBI, the timing patterns suggest that markets respond not only to the policy decision itself but also to the broader information environment leading up to it. The fact that rate-sensitive sectors move before announcements indicates that forward guidance, communication, and macroeconomic signals are already being incorporated into prices. This points to the important role of communication strategy in shaping market expectations.

From an academic standpoint, the study demonstrates the value of combining multiple methodological approaches. Each method captures a different dimension of the transmission process, and using them together provides a more complete picture than any single approach could offer.

5.5 Limitations

Like all empirical studies, this analysis comes with certain limitations.

The choice of a $[-3, +3]$ event window involves a trade-off between capturing delayed reactions and avoiding contamination from unrelated events. Different window choices may produce different results.

The study covers only six sectors, while the Indian equity market includes several others — such as Energy, Infrastructure, and Consumer Durables — that may respond differently to monetary policy.

The panel regression includes all trading days, which dilutes the policy signal since repo rate changes happen infrequently. Models focused specifically on event periods might pick up stronger effects.

Finally, the study does not distinguish between expected and unexpected components of policy changes. Prior research suggests that markets react mainly to surprises, see [Bernanke and Kuttner \(2005\)](#), and incorporating such a decomposition could add greater depth to the analysis.

5.6 Scope for Future Research

This study opens up several directions for further work. Expanding the sectoral coverage would give a more complete picture of monetary policy transmission across Indian equity markets. Incorporating measures of expected versus unexpected policy changes would allow for a more direct test of market efficiency.

Extending the analysis to volatility — using models such as GARCH — could capture how policy announcements affect risk, not just returns. A sub-period analysis comparing pre-COVID, COVID, and post-COVID phases could also reveal how the transmission mechanism has shifted over time.

This study began with a simple question — does it matter which sector you are in when the RBI speaks? The evidence suggests that it does. But the extent of that impact varies considerably across sectors, and understanding this variation is key to making sense of how monetary policy actually works its way through equity markets.

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